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Points	Rubric Criteria
25	Implementation of different lighting types - includes directional, point, spot, and area lighting. Code should run without errors and lighting differences across modes should clearly be visible.
15	Implementation of dynamic shadows, soft shadows, and ambient occlusion. Code should run without errors and the shadows should truly be dynamic and not static. Shadows should be soft and not sharp/unrealistic, with the ability to change the shadow mapping resolution.
15	Extra effort for making the scene appear realistic - one example is making the scene change over time to simulate a full day, or adding multiple dynamic elements that allow the user to change how the scene appears.
15	Report + demo video
70	TOTAL

- 1) 25 points - we implemented all different kinds of lighting (spot, point, directional, and area) in multiple different areas of our scene. The campfire uses spot, point, and area lighting, the torches use point lighting primarily, and the sun uses directional lighting to illuminate the scene as it passes by. The cabin windows and lanterns also use point light sources.
- 2) 15 points - we implemented dynamic shadows that change in length throughout the day (start long, get shorter towards noon, and get stretched out in the opposite direction towards the evening, similar to how shadows work in the real world). Shadows can be changed to be mapped to different resolutions, where the higher resolutions are the most computationally expensive, and the lower resolutions appear much fuzzier but improve performance. Soft shadows are also able to be toggled on or off. We also implemented ambient occlusion to make the scene appear less sharp and more realistic, which can be toggled on and off.
- 3) 15 points - the scene goes above and beyond to be realistic. We implemented the dynamic day and night cycle, where the sun passes by overhead during the day and when the sun sets, the moon renders and passes over the scene, with stars in the night sky using particles. The lake in the scene also captures reflections from the passing sun or moon depending on the time of the day. The scene also allows for dynamic weather by allowing

us to toggle rain, wet ground, and fog, as well as setting the intensities for each type of weather feature. We also added wildlife, such as Tung Tung Tung Sahur, fireflies, the deer, and birds, which are all animated to interact with the scene. At night, we added an aurora and shooting stars to provide some extra light to the dark nighttime scene.